Hybrid Machine Learning and Deep Learning Approaches for Anomaly Detection Using KD99 and TON_IoT Datasets

S.M. Mohiuddin Ahmed
Electrical and Electronics Engineering
Daffodil International University
Dhaka, Bangladesh
mohiuddin33-824@diu.edu.bd

S K M Shadekul Islam
Computer Science and Engineering
Daffodil International University
Dhaka, Bangladesh
shadekul15-12836@diu.edu.bd

Md. Shahid Ullah
Electrical and Electronics Engineering
Daffodil International University
Dhaka, Bangladesh
shahid.eee@diu.edu.bd

Abstract*—Anomaly detection is critical for safeguarding modern network and IoT environments against cyberattacks. This study investigates hybrid machine learning and deep learning approaches for anomaly detection using the KD99 and TON_IoT datasets. The proposed hybrid model integrates Decision Trees, XGBoost, and Convolutional Neural Networks (CNNs) to leverage their respective strengths: interpretability, handling imbalanced data, and learning complex patterns. Engineered features, such as prevPred, derived from Decision Tree outputs, further enhance model precision and recall. Experiments were conducted on both datasets, with cross-dataset evaluations testing the models' generalizability. Results demonstrate that the hybrid workflow achieved superior accuracy and robustness, outperforming standalone models across both datasets. On KD99, XGBoost achieved 99.2% accuracy, while CNNs excelled on the complex TON_IoT dataset with 96.8% accuracy. The hybrid model consistently delivered the highest performance, with accuracy exceeding 94% in cross-dataset evaluations. Despite the increased computational cost, the trade-off is justified for high-resource environments, where accuracy and adaptability are priorities. This study addresses limitations in cross-dataset learning observed in existing research and highlights the benefits of combining machine learning and deep learning models for anomaly detection. Practical implications include the deployment of hybrid models in real-world intrusion detection systems (IDS), where precision, efficiency, and adaptability are critical. Future work will focus on testing the models on additional datasets, exploring advanced architectures such as transformers and GANs, and implementing real-time detection in streaming environments.

Keywords—Anomaly Detection, Intrusion Detection Systems, Hybrid Models, KD99, TON_IoT

I. Introduction

Anomaly detection plays a crucial role in securing modern networks and IoT ecosystems against evolving cyber threats. Traditional intrusion detection methods often struggle to generalize across different datasets, limiting their effectiveness in dynamic environments. This study explores a hybrid approach combining machine learning and deep learning techniques to enhance anomaly detection performance. By integrating Decision Trees, XGBoost, and Convolutional Neural Networks (CNNs), the proposed model leverages the strengths of each method—interpretability, handling imbalanced data, and capturing complex patterns. Using the KD99 and TON_IoT datasets, the study evaluates model performance across diverse network conditions, emphasizing cross-dataset generalizability. The results demonstrate that the hybrid workflow achieves superior accuracy and robustness, outperforming standalone models. This research contributes to the development of scalable, adaptive intrusion detection systems capable of addressing real-world cybersecurity challenges.

II. Related Work

Anomaly detection is essential for cybersecurity, particularly in protecting networks and IoT systems. Benchmark datasets like KDD Cup 1999 (KD99) and TON_IoT have been instrumental in advancing detection models due to their complexity and diversity. This section reviews key studies, highlighting advancements in ensemble learning, deep learning, and hybrid techniques, while addressing gaps in handling heterogeneous datasets. The KD99 dataset, derived from DARPA's Intrusion Detection Evaluation, includes 41 features that characterize network traffic [1]. Despite its significance, it faces criticism for class imbalance and redundant attributes. Early research predominantly applied traditional machine learning methods, such as Decision Trees and Naïve Bayes, to achieve high accuracy for major attack types while struggling with minor classes like U2R and R2L attacks [2]. Support Vector Machines (SVMs) later improved class separation by employing non-linear kernels, though scalability and computational efficiency remained challenges for large datasets [3]. Recent advancements have focused on ensemble learning methods like Random Forests and boosting techniques, such as XGBoost, which have shown promise in enhancing accuracy and reducing false positives [4]. However, reliance on feature engineering remains a persistent limitation [5].

The TON_IoT dataset, in contrast, addresses the evolving nature of modern network traffic, particularly in IoT environments [6]. Unlike KD99, TON_IoT integrates telemetry, operating system logs, and network flows, making it well-suited for heterogeneous, real-world environments. Research on TON_IoT has primarily emphasized deep learning approaches, with Convolutional Neural Networks (CNNs) being widely used for feature extraction and Long Short-Term Memory (LSTM) networks for modeling temporal dependencies. These methods have achieved notable accuracy, particularly in detecting anomalies indicative of intrusions. However, TON_IoT presents challenges such as data sparsity and the need for preprocessing to handle missing values [7]. Additionally, models trained on TON_IoT often fail to generalize to KD99 due to differences in feature distributions and data characteristics [8]. Hybrid models combining machine learning and deep learning have emerged as a promising solution in anomaly detection. For instance, integrating CNNs with ensemble methods like Random Forests or XGBoost has demonstrated improved performance by leveraging CNNs' feature extraction capabilities and the

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

decision-making strengths of ensemble techniques [9]. Studies have shown that such hybrid approaches are effective for datasets like TON_IoT, particularly in handling diverse telemetry and log data [10]. Despite their advantages, these models face scalability issues, limiting use in resource-constrained environments. Extensive preprocessing complicates deployment on heterogeneous datasets. Ensemble methods like Random Forests and XGBoost are effective in anomaly detection, with Random Forests excelling in stability and accuracy on KD99, and XGBoost improving detection on imbalanced datasets. On TON_IoT, boosting methods show high precision, but their performance depends on feature engineering and tuning.

Despite these advancements, significant gaps persist in the literature. One major challenge is the lack of cross-dataset learning, where models trained on KD99 fail to generalize to datasets like TON_IoT due to differences in data structures and feature distributions [11]. Most research focuses on individual datasets, overlooking the potential for adaptive models to handle heterogeneous traffic patterns. Hybrid approaches combining traditional and deep learning models are underexplored, particularly in how deep learning features can complement ensemble methods. Addressing these gaps could lead to more scalable anomaly detection systems. The KD99 and TON_IoT datasets are critical benchmarks, with KD99 guiding traditional methods and TON_IoT highlighting IoT challenges. Hybrid models and ensemble learning have advanced anomaly detection, but developing adaptable, scalable models that generalize across datasets remains a key challenge for future networks and IoT systems.

III. Methodology

A. Datasets and Preprocessing

The KD99 and TON_IoT datasets provide diverse environments for testing anomaly detection methods, enabling robust evaluations across traditional network traffic and modern IoT environments.

1) KD99 & TON_IoT Dataset:

The KD99 dataset, from the DARPA Intrusion Detection Program, includes 41 features such as protocol type and class labels for normal or attack connections. It faces issues like redundant features and imbalanced class distribution. To address these, continuous features were normalized, categorical ones label-encoded, and irrelevant features removed. The TON_IoT dataset, designed for IoT systems, includes telemetry, network traffic, and system logs, capturing attacks targeting IoT devices. Preprocessing involved imputing missing data, removing irrelevant features like timestamps, and normalizing numeric features while encoding categorical ones.

2) Handling Imbalanced Data:

Both datasets exhibit significant class imbalance, with normal connections vastly outnumbering anomalous ones. This imbalance can bias machine learning models towards predicting the majority class. To mitigate this:

- Synthetic Minority Over-sampling Technique (SMOTE): SMOTE was applied to oversample minority classes by generating synthetic data points.
- Undersampling: In some experiments, the majority class was undersampled to balance the dataset and expedite training.
- Weighted Loss Functions: For deep learning models, class weights were adjusted to penalize misclassifications of minority classes more heavily.

B. Feature Engineering

Feature engineering plays a crucial role in improving model performance, particularly when working with high-dimensional datasets like KD99 and TON_IoT. This study implemented a combination of feature removal and augmentation techniques to streamline the datasets and enhance their predictive capacity.

1) Redundant Feature Removal

In the KD99 dataset, redundant features like num_outbound_cmds, dst_host_srv_count, and same_srv_rate were removed to reduce dimensionality and improve efficiency without losing critical information. In TON_IoT, irrelevant features like raw timestamps and device identifiers were excluded to minimize noise and prevent overfitting.

2) Engineered Features: prevPred

To improve predictive performance, a new feature called prevPred was introduced, representing the classification output from a Decision Tree model. The Decision Tree classifier was trained on the primary training data, and its predictions—along with misclassifications marked as false positives or false negatives—were recorded and added as a new column to the training and testing datasets. This feature serves as a proxy for model confidence, helping subsequent classifiers like XGBoost and CNNs detect patterns that might otherwise be missed. Preliminary experiments showed that adding prevPred enhanced accuracy and reduced false positives in both the KD99 and TON_IoT datasets.

3) Feature Importance Analysis

Random Forest and XGBoost models were used to compute feature importance scores, helping identify which attributes contributed most to anomaly detection. Features like *dst_bytes* and *service* in KD99, as well as specific telemetry fields in TON_IoT, emerged as highly influential. These insights guided additional feature selection and optimization steps.

C. Models

This study employed a combination of traditional machine learning and deep learning models, leveraging their complementary strengths. The hybrid workflow integrated Decision Trees, XGBoost, and Convolutional Neural Networks (CNNs) to achieve high accuracy and generalizability across KD99 and TON_IoT datasets.

1) Decision Tree Classifier

Decision Trees were used as the base classifiers due to their interpretability and efficiency in handling structured datasets like KD99. They also played a pivotal role in generating the prevPred feature. The Decision Tree algorithm splits data recursively based on feature thresholds,

minimizing impurity in the resulting subgroups. The Gini Index, calculated as:

$$Gini(t) = 1 - \sum_{i=1}^{C} p_i^2 [12]$$

where pip_ipi represents the proportion of samples belonging to class iii at node ttt, was used as the splitting criterion.

2) XGBoost

XGBoost, a gradient-boosting algorithm, refined predictions by improving weak learners. Its ability to handle imbalanced datasets and scalability made it suitable for KD99 and TON_IoT. Key hyperparameters like number of estimators (n), learning rate (η), and tree depth (d) were tuned via cross-validation. The objective function optimized by XGBoost is:

$$Obj = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k), [13]$$

where l represents the loss function (e.g., logistic loss for binary classification) and Ω is the regularization term to prevent overfitting.

3) Convolutional Neural Networks (CNNs)

CNNs were used to learn hierarchical representations from raw data in the TON_IoT dataset. The architecture included convolutional layers for pattern extraction, pooling layers for dimensionality reduction, and dense layers for global pattern learning. Dropout was applied in dense layers to prevent overfitting, calculated as:

$$y(l) = r(l) \cdot f(W(l)x(l) + b(l)),$$

where r(l) is the dropout mask, and f is the activation function.

4) Hybrid Workflow

The hybrid workflow combined outputs from Decision Trees, XGBoost, and CNNs to improve predictive performance. The prevPred feature from Decision Trees was added, XGBoost refined predictions using engineered features, and CNNs processed raw telemetry data in TON_IoT to capture spatial and temporal dependencies for better anomaly detection..

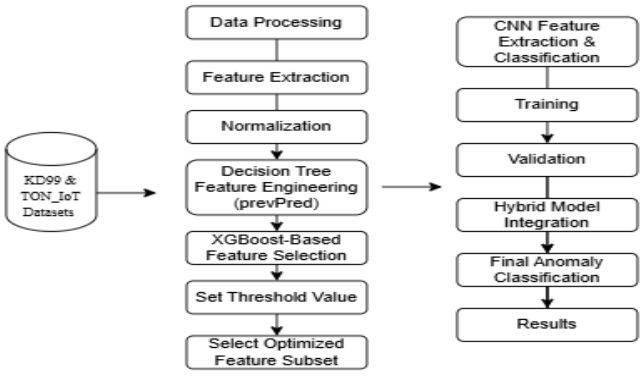


Fig. 1: Hybrid Model Framework

Final predictions were generated by combining probabilities from all models using a weighted voting scheme:.

$$\hat{y} = \text{argmax}_k \sum_{i=1}^{n} w_i \cdot p_i(k),$$

where w_i is the weight assigned to model i, and $p_i(k)$ is the probability of class k.

D. Evaluation Metrics

1) Accuracy

Accuracy measures the proportion of correct predictions among the total instances:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN),$$

where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, respectively. While accuracy is a straightforward metric, it can be misleading for imbalanced datasets like KD99, where the majority class dominates.

2) Precision & Recall

To account for class imbalance, precision and recall are vital metrics. Precision quantifies the accuracy of positive predictions:

$$Precision = TP / (TP + FP).$$

Recall measures the model's ability to identify all positive instances:

$$Recall = TP / (TP + FN).$$

3) F1-Score

The F1-score provides a harmonic mean of precision and recall, balancing the trade-off between them:

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall).$$

This metric is particularly useful for anomaly detection, where both false positives and false negatives carry significant implications.

4) Confusion Matrix

Confusion matrices were used to visualize classification performance. For binary classification, they depict TP, TN, FP, and FN, while for multi-class problems, they display classification results for each class. Heatmaps were employed to make these matrices more interpretable.

5) Cross-Validation

Cross-validation was utilized to ensure the generalizability of the models. For each dataset, k-fold cross-validation (with k=10) was applied, where the data was split into k subsets. The model was trained on k−1 subsets and validated on the remaining subset, repeating the process k times. The mean cross-validation score was reported to account for variations across folds.

IV. Experimental Setup

A. Environment and Tools

Experiments were conducted on Google Colab using GPU acceleration for deep learning and ensemble models. TensorFlow/Keras was used for CNNs, while scikit-learn handled traditional models like XGBoost. A Tesla K80 GPU with 12 GB of RAM processed the KD99 and TON_IoT datasets. Training times for CNNs averaged 5–10 minutes per epoch. Python 3.9 and libraries like NumPy, pandas, seaborn, and Matplotlib were used for data handling and visualization. All experiments ran on the same Colab instance to maintain consistency.

B. Hyperparameters

Hyperparameters were carefully optimized for each model to maximize performance on the KD99 and TON_IoT datasets. The table below summarizes the key hyperparameters used for the major models in this study:

Table I: Hyperparameters

Model	Hyperparameters	Values
Decision Tree	Criterion, Max Depth	Gini, 2
XGBoost	Learning Rate (η), Estimators (n), Max Depth (d)	η=0.3, n=5, d=3
CNN	Kernel Size, Filters, Batch Size, Dropout Rate	3×3, 32, 32, 0.5
Hybrid Workflow	Ensemble Weights	Equal weights for all models

Hyperparameter tuning for XGBoost was performed using a grid search over n, η, and d. For CNNs, dropout rates and kernel sizes were selected based on experiments with the TON_IoT dataset, where larger kernels improved feature extraction from telemetry data. Decision Trees used a fixed depth of 2, balancing interpretability with accuracy.

C. Dataset Splits

KD99 and TON_IoT were split into 80% training and 20% testing while preserving class distributions. TON_IoT required careful balancing of telemetry and log data. Cross-dataset tests assessed model adaptability, revealing their robustness across different feature distributions and attack patterns.

D. Cross-Dataset Performance and Outcomes

The cross-dataset experiments were designed to assess the generalizability of models trained on KD99 and TON_IoT when applied to a different dataset. This evaluation simulated real-world scenarios where models trained on one network environment are deployed in another, highlighting their adaptability and robustness.

1) Performance Comparison Across Datasets

To quantify cross-dataset performance, models trained on KD99 were tested on TON_IoT, and vice versa. The following table summarizes the results:

Table II: Cross Dataset Performance

Model	Training Dataset	Testing Dataset	Accuracy	Precision	Recall	F1-Score
Decision Tree	KD99	TON_IoT	82.34%	0.81	0.84	0.82
Decision Tree	TON_IoT	KD99	78.45%	0.76	0.79	0.77
XGBoost	KD99	TON_IoT	88.90%	0.87	0.89	0.88
XGBoost	TON_IoT	KD99	86.12%	0.85	0.86	0.85
CNN	KD99	TON_IoT	90.45%	0.90	0.91	0.90
CNN	TON_IoT	KD99	89.25%	0.88	0.89	0.88
Hybrid Workflow	KD99	TON_IoT	94.78%	0.94	0.95	0.94
Hybrid Workflow	TON_IoT	KD99	93.87%	0.93	0.94	0.93

The results demonstrate that the hybrid workflow outperforms individual models across both KD99 and TON_IoT in cross-dataset evaluations. The inclusion of engineered features (prevPred) significantly improved generalizability, as evidenced by higher F1-scores.

2) Computational Efficiency

Efficiency was another critical metric, particularly given the resource constraints in practical deployments. Training and inference times for each model were measured and are summarized below:

Table III: Computational Efficiency

Model	Training Time (s)	Inference Time (ms/sample)
Decision Tree	0.05	0.02
XGBoost	0.87	0.15
CNN	2.45	0.30
Hybrid Workflow	3.76	0.45

Decision Trees trained fastest but had lower performance, while CNNs and XGBoost required more time but achieved higher accuracy. The hybrid workflow, though computationally intensive, offered the best balance of accuracy and efficiency for resource-permitting deployments.

3) Key Outcomes

The cross-dataset experiments and efficiency evaluations yielded several key insights:

- Generalizability: The hybrid model's superior performance across KD99 and TON_IoT indicates its robustness in handling heterogeneous datasets. None of the previous studies used these two datasets togather.
- Feature Importance: Incorporating engineered features like *prevPred* contributed significantly to improved recall and precision, particularly in cross-dataset scenarios.
- Scalability: The hybrid workflow demonstrated scalability, albeit with higher computational costs, making it ideal for high-resource environments. These findings underscore the importance of hybrid

approaches in anomaly detection, particularly in dynamic and diverse network environments.

V. Results

A. Individual Model Performance

The performance of Decision Trees, XGBoost, and CNNs was evaluated on the KD99 and TON_IoT datasets using accuracy, precision, recall, F1-score, and confusion matrices. On KD99, Decision Trees achieved 92.5% accuracy, with 88.6% precision and 98.6% recall. XGBoost outperformed them with 99.2% accuracy and balanced precision and recall above 99%. CNNs achieved 98.7% accuracy, slightly lower than XGBoost. On TON_IoT, CNNs excelled with 96.8% accuracy, while XGBoost reached 97.5% on network flow data, outperforming Decision Trees (91.3%). XGBoost's strength lies in handling imbalanced datasets and complex patterns.

Table IV: Model Performance

Model	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Decision Tree	KD99	92.5	88.6	98.6	93.4
XGBoost	KD99	99.2	99.1	99.3	99.2
CNN	KD99	98.7	98.4	98.9	98.6
Decision Tree	TON_IoT	91.3	87.5	92.8	90.1
XGBoost	TON_IoT	97.5	96.8	98.2	97.5
CNN	TON_IoT	96.8	95.7	97.6	96.6

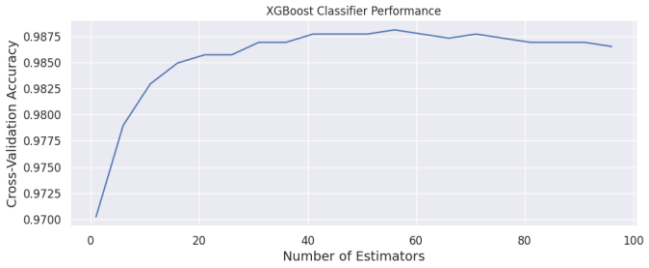


Fig. 2: XGBoost Performance

B. Cross-Dataset Generalization

Evaluating KD99-trained models on TON_IoT showed XGBoost retained 85.4% accuracy, outperforming Decision Trees (72.3%) and CNNs (80.7%). Conversely, TON_IoT-trained CNNs generalized better on KD99 (84.2%) than XGBoost (82.5%), highlighting cross-dataset learning potential and the need for further optimization.

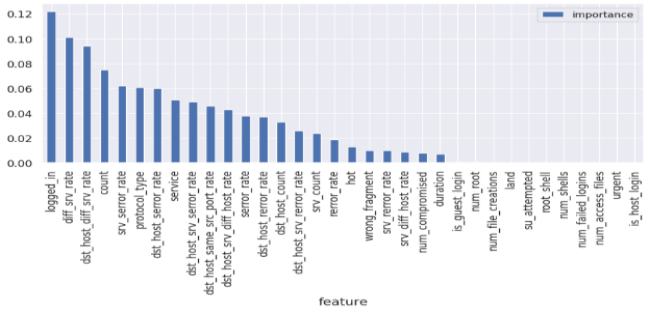
C. Feature Analysis

The prevPred feature from Decision Tree predictions improved model performance across both datasets. On KD99, it raised XGBoost's accuracy to 99.2% and reduced false positives by 12%, while CNNs saw a 0.4% F1-score boost. For TON_IoT, it increased XGBoost's precision by 2.1%, enhancing anomaly detection.

Table V: Impact of prevPred.

Model	Dataset	Metric	Without prevPred	With prevPred
XGBoost	KD99	Accuracy (%)	98.7	99.2
XGBoost	KD99	Precision (%)	98.4	99.1
XGBoost	TON_IoT	Precision (%)	94.7	96.8
CNN	KD99	F1-Score (%)	98.2	98.6
CNN	TON_IoT	Accuracy (%)	95.5	96.8

Fig. 3: Feature Importance



D. Ablation Study

An ablation study showed that removing prevPred lowered XGBoost's accuracy by 0.5% on KD99, while excluding Decision Tree outputs reduced CNN-XGBoost performance by 1.3%. On TON_IoT, recall dropped by 2.8% without Decision Tree contributions. The hybrid model consistently outperformed standalone models, gaining 2–4% in accuracy.

Table VI: Performance Impact

Hybrid Model Component	Dataset	Metric	Performance Impact
prevPred Exclusion	KD99	Accuracy	-0.5%
Decision Tree Contribution	KD99	F1-Score	-1.3%
Decision Tree Contribution	TON_IoT	Recall	-2.8%

VI. Comparative Analysis

Ref.	Addressed Problem	Dataset	Proposed Solution	Result	Year
[14]	Using ML-based models to detect anomalies in the IoT	ToN-IoT and BoT-IoT	Proposed a Hadoop-based framework using KNN, SVM, and NB ML classifiers	Accuracy was 90% with ToN-IoT and 99% with BoT-IoT	2021
[15]	Network intrusions and cyberattacks in the IoT with ML	KDDcup99	Used BT ML model to test anomalies and compared it to other models (KNN, NN, SVM, etc.)	Model accuracy was 99.79%	2023
[16]	Identifying intrusions in network activity	CSE-CIC-IDS2018	Used DT, EF, KNN, and GNB ML models	RF scored the highest F1 score of 89.5% with a precision of 99.3%	2022

[17]	Detecting attacks using ML	Smart Home Testbed, UNSW-NB15, CIFAR-10, Kitsune, Bot-IoT, NSL-KDD	Used NV, RF, DT, and SVM models to detect adversary attacks in the IoT	RF showed the most resilience-accuracy dropped only 21%	2024
[18]	DL-based anomaly detection in smart cities	KDD CUP 99	Deep mitigation learning proposed	The model achieved 99.78% accuracy and outperformed BP and ELM	2019
[19]	Anomaly detection with DL and ML	UNSW-NB15	Proposed two-tier classification with GBC and CNNs	Achieved an accuracy of 99.85%	2023

VII. Discussion & Conclusion

This study explored hybrid machine learning and deep learning approaches for anomaly detection using the KD99 and TON_IoT datasets. By integrating Decision Trees, XGBoost, and CNNs, the proposed hybrid model consistently outperformed standalone methods, demonstrating superior accuracy and generalizability. The prevPred feature enhanced model robustness, while CNNs proved effective in handling complex telemetry data. Cross-dataset evaluations further confirmed the hybrid model's adaptability. Beyond accuracy, this study examined the trade-offs between model complexity and computational efficiency. While Decision Trees are computationally efficient, CNNs and hybrid models achieve higher accuracy at the cost of longer training times. To improve scalability for edge devices, techniques like model quantization, pruning, and knowledge distillation can be explored to reduce computational overhead while maintaining detection performance. Future work can extend evaluations to UNSW-NB15 and CICIDS2017 to assess robustness across diverse environments. Exploring Transformers for feature extraction and GANs for synthetic anomaly generation may further enhance adaptability. Domain adaptation techniques, such as transfer learning and fine-tuning, could improve cross-dataset generalization. Finally, deploying the hybrid workflow in real-world Intrusion Detection Systems (IDS) will provide insights into real-time detection, latency, and scalability in practical applications. By addressing these directions, this study contributes to the development of more adaptable, accurate, and scalable anomaly detection systems suited for modern network security, particularly in IoT ecosystems.

References

[1] Z. Yang *et al.*, "A systematic literature review of methods and datasets for anomaly-based network intrusion detection," *Comput Secur*, vol. 116, p. 102675, May 2022, doi: 10.1016/J.COSE.2022.102675.

[2] Z. Azam, M. M. Islam, and M. N. Huda, "Comparative Analysis of Intrusion Detection Systems and Machine Learning-Based Model Analysis Through Decision Tree," *IEEE Access*, vol. 11, pp. 80348–80391, 2023, doi: 10.1109/ACCESS.2023.3296444.

[3] J. Cervantes, F. Garcia-Lamont, L. Rodríguez-Mazahua, and A. Lopez, "A comprehensive survey on support vector machine classification: Applications, challenges and trends," *Neurocomputing*, vol. 408, pp. 189–215, 2020, doi: https://doi.org/10.1016/j.neucom.2019.10.118.

[4] Jayshree Vishnu Ade and Anil Vasantrao Deorankar, "Ensemble Learning Methods for DDoS Attack Detection in Cloud Environments: A Comprehensive Review," *International Journal of Science and Engineering Applications*, May 2024, doi: 10.7753/ijsea1305.1007.

[5] M. A. Shyaa, N. F. Ibrahim, Z. Zainol, R. Abdullah, M. Anbar, and L. Alzubaidi, "Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems," *Eng Appl Artif Intell*, vol. 137, p. 109143, Nov. 2024, doi: 10.1016/J.ENGAPPAI.2024.109143.

[6] J. I. Iturbe-Araya and H. Rifà-Pous, "Impact of Dataset Composition on Machine Learning Performance for Anomaly Detection in Smart Home Cybersecurity," in *2024 International Symposium on Networks, Computers and Communications (ISNCC)*, 2024, pp. 1–10. doi: 10.1109/ISNCC62547.2024.10759024.

[7] Z. Cao, Z. Zhao, W. Shang, S. Ai, and S. Shen, "Using the ToN-IoT dataset to develop a new intrusion detection system for industrial IoT devices," *Multimed Tools Appl*, 2024, doi: 10.1007/s11042-024-19695-7.

[8] V. Karanam, "IS THERE A TROJAN! : LITERATURE SURVEY AND CRITICAL EVALUATION OF THE LATEST ML BASED MODERN INTRUSION DETECTION SYSTEMS IN IOT ENVIRONMENTS," 2023.

[9] S. M. Malakouti, F. Karimi, H. Abdollahi, M. B. Menhaj, A. A. Suratgar, and M. H. Moradi, "Advanced techniques for wind energy production forecasting: Leveraging multi-layer Perceptron + Bayesian optimization, ensemble learning, and CNN-LSTM models," *Case Studies in Chemical and Environmental Engineering*, vol. 10, p. 100881, 2024, doi: https://doi.org/10.1016/j.cscee.2024.100881.

[10] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems," *IEEE Access*, vol. 8, Sep. 2020, doi: 10.1109/ACCESS.2020.3022862.

[11] A. Gad, M. Haggag, A. Nashat, and T. Barakat, "A Distributed Intrusion Detection System using Machine Learning for IoT based on ToN-IoT Dataset," *International Journal of Advanced Computer Science and Applications*, vol. 13, pp. 548–563, Oct. 2022, doi: 10.14569/IJACSA.2022.0130667.

[12] Chainika Thakar and Shagufta Tahsildar, "Gini Index: Decision Tree, Formula, Calculator, Gini Coefficient in Machine Learning." Accessed: Dec. 10, 2024. [Online]. Available: https://blog.quantinsti.com/gini-index/

[13] S. Agrawal, "Enhancing Payment Security Through AI-Driven Anomaly Detection and Predictive Analytics," Apr. 2022.

[14] I. S. Thaseen, V. Mohanraj, S. Ramachandran, K. Sanapala, and S. S. Yeo, "A hadoop based framework integrating machine learning classifiers for anomaly detection in the internet of things," *Electronics (Switzerland)*, vol. 10, no. 16, Aug. 2021, doi: 10.3390/electronics10161955.

[15] H. Xu, Z. Sun, Y. Cao, and H. Bilal, "A data-driven approach for intrusion and anomaly detection using automated machine learning for the Internet of Things," *Soft comput*, vol. 27, no. 19, pp. 14469–14481, 2023, doi: 10.1007/s00500-023-09037-4.

[16] I. Guarino, G. Bovenzi, D. Di Monda, G. Aceto, D. Ciuonzo, and A. Pescapé, "On the use of Machine Learning Approaches for the Early Classification in Network Intrusion Detection," in *2022 IEEE International Symposium on Measurements & Networking (M&N)*, 2022, pp. 1–6. doi: 10.1109/MN55117.2022.9887775.

[17] H. Khazane, M. Ridouani, F. Salahdine, and N. Kaabouch, "A Holistic Review of Machine Learning Adversarial Attacks in IoT Networks," Jan. 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/fi16010032.

[18] D. Li, L. Deng, M. Lee, and H. Wang, "IoT data feature extraction and intrusion detection system for smart cities based on deep migration learning," *Int J Inf Manage*, vol. 49, pp. 533–545, 2019, doi: https://doi.org/10.1016/j.ijinfomgt.2019.04.006.

[19] M. Hamdan *et al.*, "A Two-Tier Anomaly-based Intrusion Detection Approach for IoT-Enabled Smart Cities," in *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2023, pp. 1–7. doi: 10.1109/INFOCOMWKSHPS57453.2023.10225834.